OS-ORANGE
LAB REPORT

PES2UG24CS102
AYUSHMAN DEY

Github link:-
https://github.com/ANDYGAMER4321/PES2UG24CS102-pes-vcs

PHASE-I

1A:

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B:

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
```

PHASE-II

2A:

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B:

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ xxd .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1a
f8e11ec4f476
00000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
00000010: 6520 636f 6e74 656e 740a                 e content.
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$
```

PHASE-III

3A:

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ ./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  README.md
  untracked:  test_objects
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h
```

3B:

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776472006 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776472006 6 file2.txt
```

PHASE-IV

4A:

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ ./pes commit -m "Initial commit with file1 and file2"
./pes log
Committed: 6130a695148e... Initial commit with file1 and file2
commit 6130a695148e57d2540104611fa7464ee28044e9ee52774dceb8be2b78241da7
Author: Ayushman Dey PES2UG24CS102
Date:   1776472233

    Initial commit with file1 and file2
```

4B:

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/61/30a695148e57d2540104611fa7464ee28044e9ee52774dceb8be2b78241da7
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2
.pes/refs/heads/main
```

4C:

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
6130a695148e57d2540104611fa7464ee28044e9ee52774dceb8be2b78241da7
ref: refs/heads/main
```

TEST-INTEGRATION:-

```
andy@andy-Lenovo-V15-G2-ALC-Ua:~/Desktop/COLLEGE/OS/orange2/PES2UG24CS102-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 059d6a8c7ec7... Initial commit

Log after first commit:
commit 059d6a8c7ec70b158fb6f931b20b6eb16aba4ff685db09ca1c8a0890bb113341
Author: Ayushman Dey PES2UG24CS102
Date:   1776472346

    Initial commit


--- Second Commit ---
Committed: 50247fc25221... Update file.txt

--- Third Commit ---
Committed: 07a9c3fb8f2f... Add farewell
```

```
--- Full History ---
commit 07a9c3fb8f2f149387c671a72ae0b462200ded79a79b7995ec423377c8aa237e
Author: Ayushman Dey PES2UG24CS102
Date:   1776472346

    Add farewellA◆

commit 50247fc25221c0303e951c87f890ccd7b8ff5e2f49c10ca981596aeda70987da
Author: Ayushman Dey PES2UG24CS102
Date:   1776472346

    Update file.txt

commit 059d6a8c7ec70b158fb6f931b20b6eb16aba4ff685db09ca1c8a0890bb113341
Author: Ayushman Dey PES2UG24CS102
Date:   1776472346

    Initial committial cQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
07a9c3fb8f2f149387c671a72ae0b462200ded79a79b7995ec423377c8aa237e

--- Object Store ---
Objects created:
10
.pes/objects/05/9d6a8c7ec70b158fb6f931b20b6eb16aba4ff685db09ca1c8a0890bb113341
.pes/objects/07/a9c3fb8f2f149387c671a72ae0b462200ded79a79b7995ec423377c8aa237e
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/50/247fc25221c0303e951c87f890ccd7b8ff5e2f49c10ca981596aeda70987da
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

PHASE-V:

Q5.1: To implement pes checkout <branch>, the system must update .pes/HEAD to point to the new branch ref, retrieve the commit hash from that ref, and walk its tree object. The working directory must then be updated to match the blobs in that tree. This is complex because it requires safely deleting files not in the new tree and overwriting others without losing uncommitted user data.

Q5.2: A "dirty" directory is detected by comparing the current file hashes in the working directory against the hashes in the .pes/index. If they differ, the file has unstaged changes. If the file also

differs between the current branch and the target branch, checkout must refuse to prevent overwriting those changes.

Q5.3: In a "Detached HEAD" state, commits are created normally but are not linked to a branch reference. If the user checks out another branch, these floating commits become "dangling" and unreachable. A user could recover them by finding the commit hash in the object store and manually creating a new branch ref for it.

PHASE-VI

Q6.1: A mark-and-sweep algorithm is used. Starting from all refs in .pes/refs/heads/, the system "marks" all reachable commits, trees, and blobs using a hash set. For 100,000 commits, it would need to visit every unique object in that history (likely several hundred thousand). Any object in .pes/objects/ not in the "marked" set is deleted.

Q6.2: Concurrent GC is dangerous because a commit operation might calculate a hash and prepare to reference an object that GC just identified as "unreachable." GC could delete a new blob before the commit points to it. Real Git avoids this by only deleting unreachable objects older than a certain age (grace period).